

Methodology

- **Data Cleaning:** Regex extraction `r"(\d+\.\d*)"` converts mixed-unit strings such as "1197 cc" into numeric values; currency fields are stripped of "Rs." and commas before conversion.
- **Critical Rows:** Rows missing Ex-Showroom_Price, Displacement, Power, or derived Mileage are removed.
- **Mileage Fallback:** If mileage is missing, it is estimated as

$$\text{Mileage} = \text{clip}\left(25 - \frac{CC}{200}, 5, 25\right)$$

- **Stage 1 – Classification:** Random Forest predicts **Low/Mid/High** budget class; CART is retained as an interpretable comparison model.
- **Stage 2 – Recommendation Ranking:** Candidate cars are ranked using similarity between user preference vectors and vehicle specification vectors.
- **Important Separation:** Classification accuracy measures budget prediction; match score measures recommendation fit. They are reported separately.